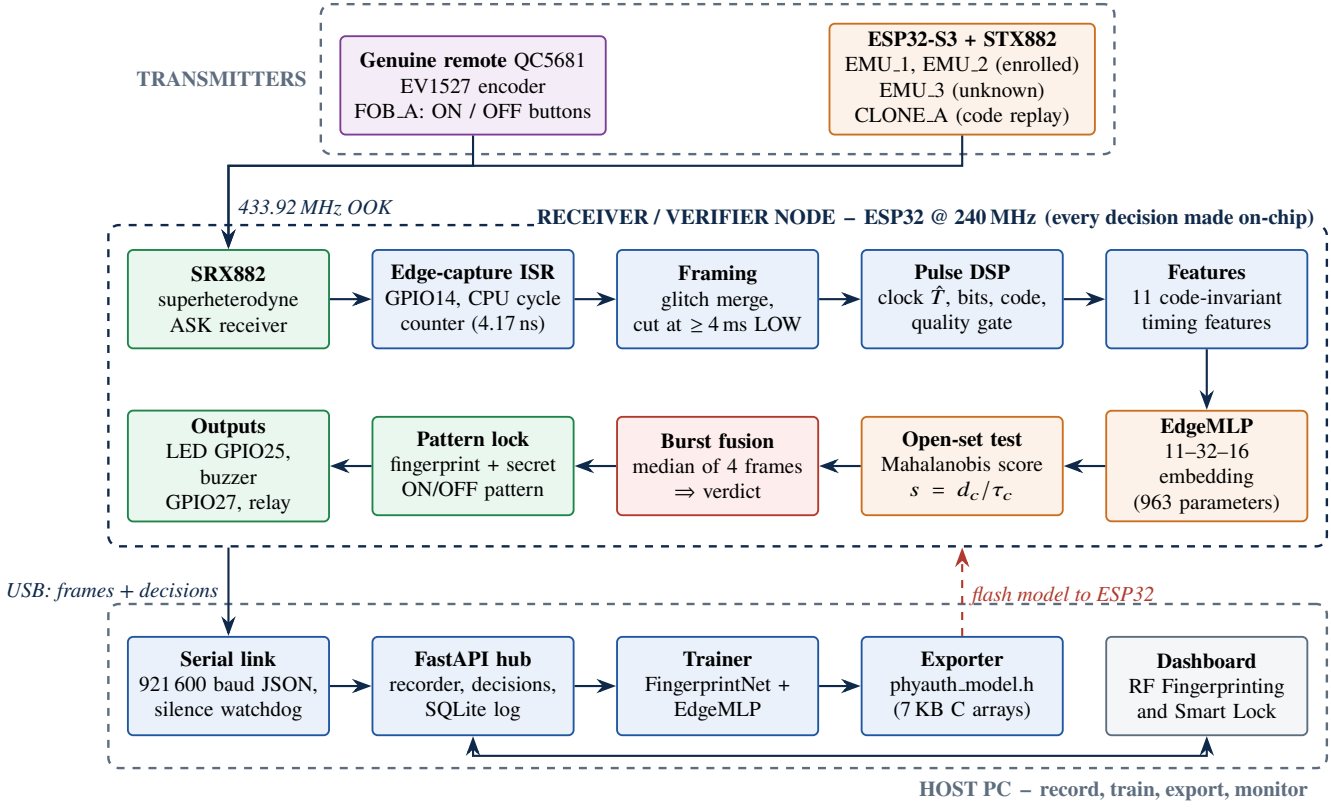

TRANSMITTERS
Genuine remote QC5681
EV1527 encoder
FOB_A: ON / OFF buttons
ESP32-S3 + STX882
EMU_1, EMU_2 (enrolled)
EMU_3 (unknown)
CLONE_A (code replay)
433.92 MHz OOK
RECEIVER / VERIFIER NODE – ESP32 @ 240 MHz (every decision made on-chip)
SRX882
superheterodyne
ASK receiver
Edge-capture ISR
GPIO14, CPU cycle
counter (4.17 ns)
Framing
glitch merge,
cut at ≥ 4 ms LOW
Pulse DSP
clock $\hat{T}$, bits, code,
quality gate
Features
11 code-invariant
timing features
EdgeMLP
11–32–16
embedding
(963 parameters)
Open-set test
Mahalanobis score
$s = d_c/\tau_c$
Burst fusion
median of 4 frames
⇒ verdict
Pattern lock
fingerprint + secret
ON/OFF pattern
Outputs
LED GPIO25,
buzzer
GPIO27, relay
USB: frames + decisions
flash model to ESP32
Serial link
921 600 baud JSON,
silence watchdog
FastAPI hub
recorder, decisions,
SQLite log
Trainer
FingerprintNet +
EdgeMLP
Exporter
phyauth_model.h
(7 KB C arrays)
Dashboard
RF Fingerprinting
and Smart Lock
HOST PC – record, train, export, monitor